Numpy

In [4]:
```
import numpy as np
```

In [5]:
```
np._version_
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[5], line 1
----> 1 np._version_

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute '_version_'
```

In [ ]:
```
np.__version__
```

In [ ]:
```
my_list=[0,1,2,3,4,5]
my_list
```

In [ ]:
```
type(my_list)
```

In [ ]:
```
arr = np.array(my_list)
arr
```

In [ ]:
```
type(arr)
```

In [ ]:
```
arr
```

In [ ]:
```
np.arange(arr)
```

In [ ]:
```
np.arange(10)
```

In [ ]:
```
np.arange(10,20)
```

In [ ]:
```
np.arange(10,20,3)
```

In [ ]:
```
np.zeros(5)
```

In [ ]:
```
np.zeros(5, dtype=int)
```

In [ ]:
```
np.zeros(2,2)
```

In [ ]:
```
np.zeros((2,2))
```

In [ ]:
```
np.zeros((2,2), dtype=int)
```

In [ ]:
```
np.zeros((5,10), dtype=int)
```

In [ ]:
```
np.ones(10)
```

In [ ]:
```
np.ones((3,3),dtype=int)
```

In [ ]:
```
np.twos(3)
```

In [ ]:
```
np.fours(3)
```

In [ ]:
```
rand(4)
```

In [ ]:
```
random.rand(4)
```

In [ ]:
```
np.random.rand(4)
```

In [ ]:
```
np.random.rand(4,4)
```

In [ ]:
```
np.random.rand(3,2)
```

In [ ]:
```
np.random.randint(3)
```

In [ ]:
```
np.random.randint(10,15)
```

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In [6]:
```
np.random.randint(1,5,3)
```

Out[6]:
```
array([3, 2, 2], dtype=int32)
```

In [10]:
```
np.random.randint(1,5,4)
```

Out[10]:
```
array([4, 4, 1, 2], dtype=int32)
```

In [11]:
```
np.random.randint(1,5,2)
```

Out[11]:
```
array([2, 3], dtype=int32)
```

In [12]:
```
np.random.randint(5,2,3)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[12], line 1
----> 1 np.random.randint(5,2,3)

File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()

File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers._rand_int32()

ValueError: low >= high
```

In [15]:
```
np.random.randint(-5,2,3)
```

Out[15]:
```
array([-4, -1, -2], dtype=int32)
```

In [17]:
```
np.random.randint(2,5,3)
```

Out[17]:
```
array([4, 4, 2], dtype=int32)
```

In [21]:
```
np.random.randint(10,40,(10,10))
```

Out[21]:
```
array([[31, 38, 18, 25, 34, 36, 39, 16, 19, 13],
       [20, 25, 24, 10, 15, 29, 36, 34, 33, 36],
       [26, 11, 38, 17, 12, 24, 14, 23, 23, 38],
       [27, 18, 26, 27, 11, 31, 30, 21, 35, 27],
       [11, 33, 18, 10, 35, 19, 31, 24, 38, 26],
       [25, 15, 32, 22, 17, 26, 21, 28, 35, 31],
       [27, 14, 29, 11, 33, 22, 17, 36, 25, 25],
       [24, 22, 15, 28, 26, 22, 17, 30, 26, 16],
       [21, 12, 27, 12, 12, 21, 37, 17, 11, 11],
       [30, 11, 21, 36, 25, 23, 35, 14, 28, 29]], dtype=int32)
```

In [29]:
```
ri=np.random.randint(10,40,(10,10))
ri
```

Out[29]:
```
array([[19, 21, 14, 17, 18, 17, 21, 11, 38, 27],
       [32, 31, 26, 39, 22, 10, 28, 30, 34, 23],
       [16, 12, 33, 21, 30, 13, 20, 11, 20, 19],
       [39, 28, 10, 25, 22, 14, 18, 15, 32, 36],
       [18, 28, 21, 39, 12, 11, 28, 23, 39, 14],
       [33, 19, 16, 29, 21, 16, 26, 39, 28, 31],
       [15, 29, 35, 18, 28, 11, 21, 10, 29, 18],
       [25, 11, 11, 37, 21, 32, 28, 28, 12, 16],
       [32, 13, 13, 33, 16, 35, 15, 13, 37, 33],
       [24, 11, 28, 24, 18, 17, 38, 17, 24, 23]], dtype=int32)
```

In [30]:
```
len(ri)
```

Out[30]:
```
10
```

In [31]:
```
np.random.randint(10,40,(3,5))
```

Out[31]:
```
array([[17, 35, 38, 13, 18],
       [12, 35, 25, 26, 35],
       [23, 23, 26, 23, 28]], dtype=int32)
```

In [35]:
```
ri=np.random.randint(10,40,(3,5))
ri
```

Out[35]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

In [36]:
```
len(ri)
```

Out[36]:
```
3
```

In [37]:
```
ri.shape
```

Out[37]:
```
(3, 5)
```

In [38]:
```
id(ri)
```

Out[38]:
```
2150940726864
```

In [39]:
```
ri
```

Out[39]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

slicing in numpy

In [40]:
```
ri[:]
```

Out[40]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

In [42]:
```
ri[1]
```

Out[42]:
```
array([35, 12, 39, 14, 16], dtype=int32)
```

In [43]:
```
ri[0]
```

Out[43]:
```
array([38, 21, 19, 23, 35], dtype=int32)
```

In [44]:
```
ri[4]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[44], line 1
----> 1 ri[4]

IndexError: index 4 is out of bounds for axis 0 with size 3
```

In [45]:
```
ri[0:3]
```

Out[45]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

In [46]:
```
ri[0:4]
```

Out[46]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

In [52]:
```
ri
```

Out[52]:
```
array([[38, 21, 19, 23, 35],
       [35, 12, 39, 14, 16],
       [29, 23, 24, 17, 24]], dtype=int32)
```

In [53]:
```
ri_1=np.random.randint(10,40,(10,10))
ri_1
```

Out[53]:
```
array([[36, 12, 34, 38, 14, 23, 37, 34, 24, 33],
       [11, 34, 25, 10, 35, 12, 29, 17, 15, 20],
       [31, 27, 13, 16, 26, 30, 19, 20, 25, 11],
       [15, 20, 34, 37, 23, 31, 19, 17, 24, 12],
       [16, 24, 30, 27, 26, 21, 38, 23, 32, 32],
       [38, 20, 18, 11, 35, 34, 16, 10, 16, 16],
       [25, 24, 25, 13, 37, 13, 22, 22, 32, 37],
       [38, 28, 31, 12, 39, 28, 37, 35, 21, 30],
       [25, 21, 28, 30, 39, 34, 29, 18, 31, 11],
       [17, 36, 35, 21, 38, 23, 28, 13, 17, 32]], dtype=int32)
```

In [54]:
```
ri_1.shape
```

Out[54]:
```
(10, 10)
```

In [55]:
```
ri_1=np.random.randint(10,40,(10,10))
print(ri_1)
```

```
[[22 35 30 39 13 31 38 28 32 24]
 [26 20 23 22 34 18 11 11 25 31]
 [35 37 17 15 10 18 22 34 24 32]
 [31 29 37 29 15 20 24 34 26 25]
 [39 18 10 25 12 23 26 33 13 31]
 [12 39 23 23 22 34 14 36 31 34]
 [36 28 21 26 36 15 26 39 10 21]
 [37 37 35 29 39 24 27 14 14 29]
 [11 26 38 17 10 33 23 39 34 29]
 [36 35 36 21 36 30 18 19 32 39]]
```

In [56]:
```
ri_1
```

Out[56]:
```
array([[22, 35, 30, 39, 13, 31, 38, 28, 32, 24],
       [26, 20, 23, 22, 34, 18, 11, 11, 25, 31],
       [35, 37, 17, 15, 10, 18, 22, 34, 24, 32],
       [31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31],
       [12, 39, 23, 23, 22, 34, 14, 36, 31, 34],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [37, 37, 35, 29, 39, 24, 27, 14, 14, 29],
       [11, 26, 38, 17, 10, 33, 23, 39, 34, 29],
       [36, 35, 36, 21, 36, 30, 18, 19, 32, 39]], dtype=int32)
```

In [58]:
```
ri_1[3:8]
```

Out[58]:
```
array([[31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31],
       [12, 39, 23, 23, 22, 34, 14, 36, 31, 34],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [37, 37, 35, 29, 39, 24, 27, 14, 14, 29]], dtype=int32)
```

In [60]:
```
print(ri_1[3,8])
```

```
26
```

In [61]:
```
ri_1
```

Out[61]:
```
array([[22, 35, 30, 39, 13, 31, 38, 28, 32, 24],
       [26, 20, 23, 22, 34, 18, 11, 11, 25, 31],
       [35, 37, 17, 15, 10, 18, 22, 34, 24, 32],
       [31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31],
       [12, 39, 23, 23, 22, 34, 14, 36, 31, 34],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [37, 37, 35, 29, 39, 24, 27, 14, 14, 29],
       [11, 26, 38, 17, 10, 33, 23, 39, 34, 29],
       [36, 35, 36, 21, 36, 30, 18, 19, 32, 39]], dtype=int32)
```

In [62]:
```
ri_1[-5]
```

Out[62]:
```
array([12, 39, 23, 23, 22, 34, 14, 36, 31, 34], dtype=int32)
```

In [63]:
```
ri_1[-5,-3]
```

Out[63]:
```
np.int32(36)
```

In [64]:
```
print(ri_1[-5,-3])
```

```
36
```

In [65]:
```
ri_1[::3]
```

Out[65]:
```
array([[22, 35, 30, 39, 13, 31, 38, 28, 32, 24],
       [31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [36, 35, 36, 21, 36, 30, 18, 19, 32, 39]], dtype=int32)
```

In [67]:
```
ri_1[::-1]
```

Out[67]:
```
array([[36, 35, 36, 21, 36, 30, 18, 19, 32, 39],
       [11, 26, 38, 17, 10, 33, 23, 39, 34, 29],
       [37, 37, 35, 29, 39, 24, 27, 14, 14, 29],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [12, 39, 23, 23, 22, 34, 14, 36, 31, 34],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31],
       [31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [35, 37, 17, 15, 10, 18, 22, 34, 24, 32],
       [26, 20, 23, 22, 34, 18, 11, 11, 25, 31],
       [22, 35, 30, 39, 13, 31, 38, 28, 32, 24]], dtype=int32)
```

In [68]:
```
ri_1[::-5]
```

Out[68]:
```
array([[36, 35, 36, 21, 36, 30, 18, 19, 32, 39],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31]], dtype=int32)
```

In [69]:
```
ri_1[2]
```

Out[69]:
```
array([35, 37, 17, 15, 10, 18, 22, 34, 24, 32], dtype=int32)
```

In [71]:
```
print(ri_1[2])
```

```
[35 37 17 15 10 18 22 34 24 32]
```

In [72]:
```
print(ri_1[:,2])
```

```
[30 23 17 37 10 23 21 35 38 36]
```

In [73]:
```
ri_1
```

Out[73]:
```
array([[22, 35, 30, 39, 13, 31, 38, 28, 32, 24],
       [26, 20, 23, 22, 34, 18, 11, 11, 25, 31],
       [35, 37, 17, 15, 10, 18, 22, 34, 24, 32],
       [31, 29, 37, 29, 15, 20, 24, 34, 26, 25],
       [39, 18, 10, 25, 12, 23, 26, 33, 13, 31],
       [12, 39, 23, 23, 22, 34, 14, 36, 31, 34],
       [36, 28, 21, 26, 36, 15, 26, 39, 10, 21],
       [37, 37, 35, 29, 39, 24, 27, 14, 14, 29],
       [11, 26, 38, 17, 10, 33, 23, 39, 34, 29],
       [36, 35, 36, 21, 36, 30, 18, 19, 32, 39]], dtype=int32)
```

In [75]:
```
ri_1[:,8]
```

Out[75]:
```
array([32, 25, 24, 26, 13, 31, 10, 14, 34, 32], dtype=int32)
```

In [76]:
```
ri_1[:,-8]
```

Out[76]:
```
array([30, 23, 17, 37, 10, 23, 21, 35, 38, 36], dtype=int32)
```

In [77]:
```
ri_1[2:6,5:8]
```

Out[77]:
```
array([[18, 22, 34],
       [20, 24, 34],
       [23, 26, 33],
       [34, 14, 36]], dtype=int32)
```

In [78]:
```
ri_1[0:3,0:3]
```

Out[78]:
```
array([[22, 35, 30],
       [26, 20, 23],
       [35, 37, 17]], dtype=int32)
```

In [80]:
```
np.arange(1,13)
```

Out[80]:
```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

In [81]:
```
np.arange(1,13).reshape(3,4)
```

Out[81]:
```
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12]])
```

In [82]:
```
np.arange(1,13).reshape(5,4)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[82], line 1
----> 1 np.arange(1,13).reshape(5,4)

ValueError: cannot reshape array of size 12 into shape (5,4)
```

In [84]:
```
np.arange(1,13).reshape(4,3)
```

Out[84]:
```
array([[ 1,  2,  3],
       [ 4,  5,  6],
       [ 7,  8,  9],
       [10, 11, 12]])
```

In [85]:
```
np.arange(1,13).reshape(2,6)
```

Out[85]:
```
array([[ 1,  2,  3,  4,  5,  6],
       [ 7,  8,  9, 10, 11, 12]])
```

In [86]:
```
np.arange(1,13).reshape(1,12)
```

Out[86]:
```
array([[ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12]])
```

Numpy masking or filter

In [89]:
```
mat=np.arange(0,100).reshape(10,10)
mat
```

Out[89]:
```
array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
       [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
       [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
       [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
       [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
       [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
       [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
       [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
       [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

In [90]:
```
id(mat)
```

Out[90]:
```
2150948507632
```

In [91]:
```
mat.shape
```

Out[91]:
```
(10, 10)
```

In [92]:
```
mat<50
```

Out[92]:
```
array([[ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [ True,  True,  True,  True,  True,  True,  True,  True,  True,
         True],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False],
       [False, False, False, False, False, False, False, False, False,
        False]])
```

In [93]:
```
mat[mat<50]
```

Out[93]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

In [95]:
```
mat[mat<=50]
```

Out[95]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

In [96]:
```
mat[mat==50]
```

Out[96]: array([50])

In [97]:
```
mat[mat!=50]
```

Out[97]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
       34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 51,
       52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68,
       69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85,
       86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

In [98]:
```
mat[mat>50]
```

Out[98]:
```
array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
       68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
       85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

In [ ]:
```
np.
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: